

Decentralized Stability Certificates in IBR-Dominated Grids

The Role of the Network State

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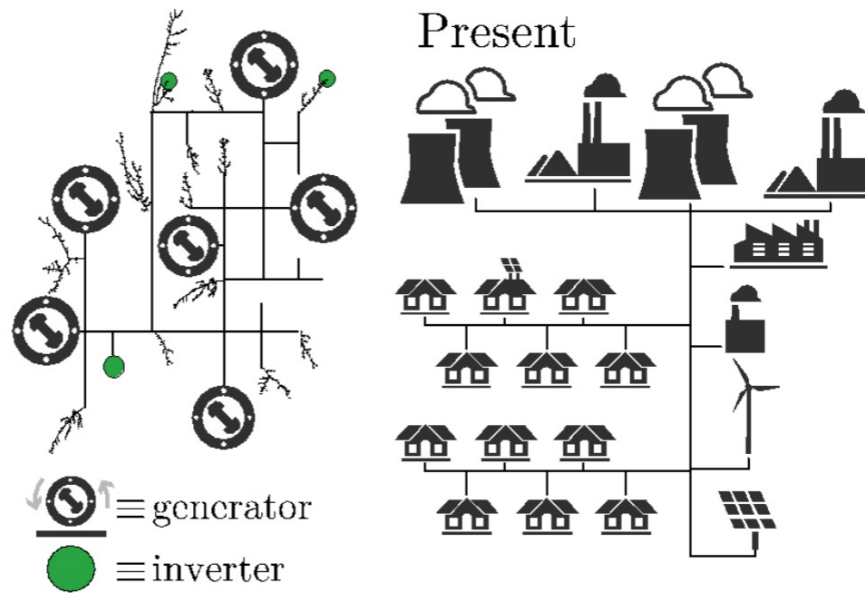
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PowerUp 2026

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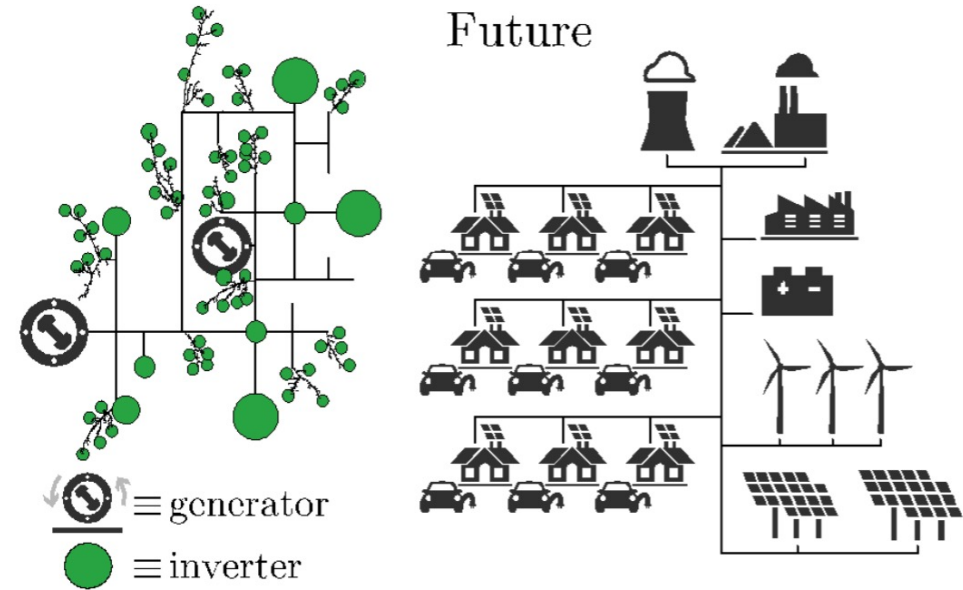


Motivation



Conventional grids

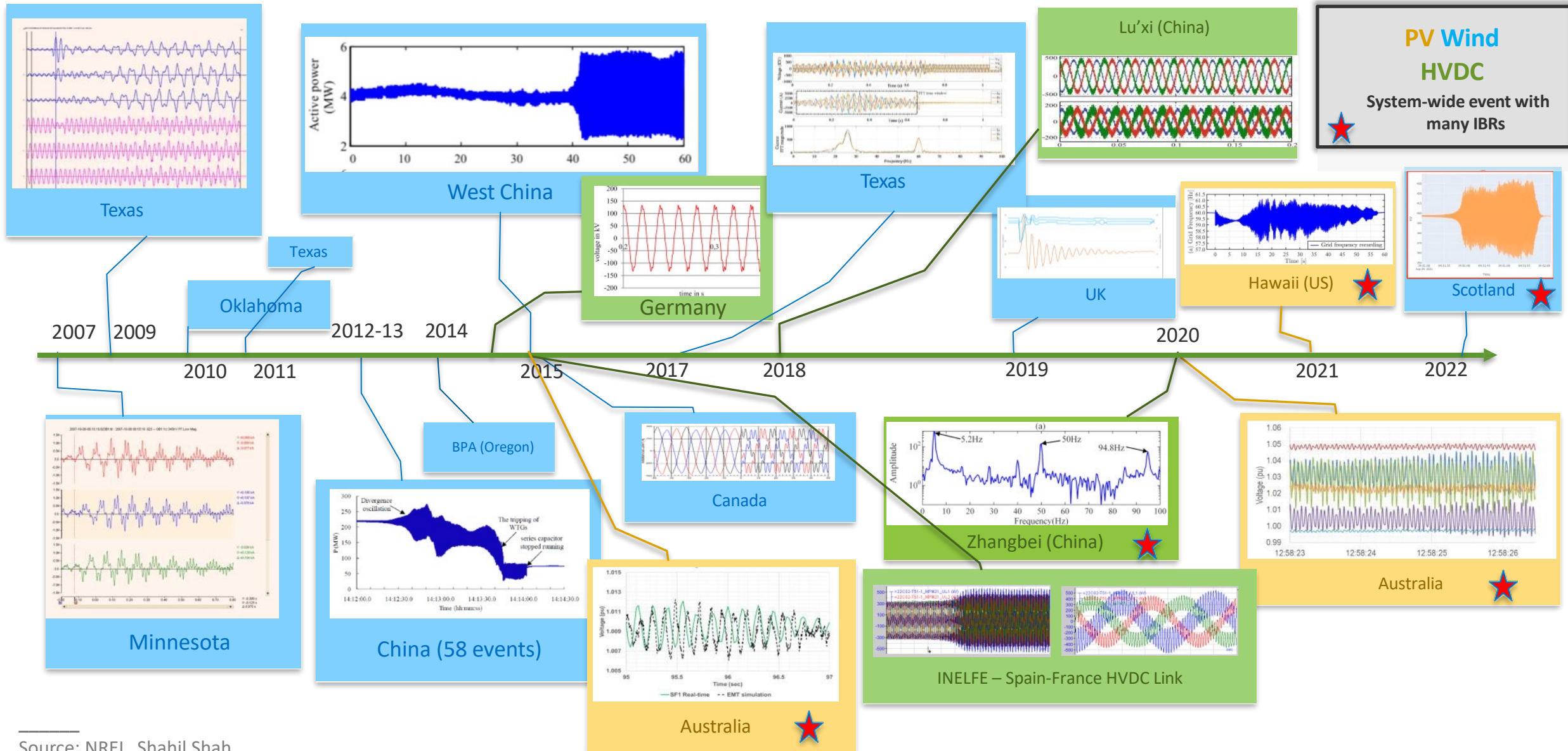
- Centralized, dispatchable generation
- Inherent inertial response
- Stronger voltage support



IBR-rich grids

- Distributed, variable generation
- Control-based frequency responses
- Control-dependent voltage support

Oscillation Events Involving IBRs



Source: NREL, Shahil Shah

[TPWRS 23] Cheng et al., "Real-World Sub-synchronous Oscillation Events in Power Grids With High Penetrations of Inverter-Based Resources," IEEE TPWRS 2023

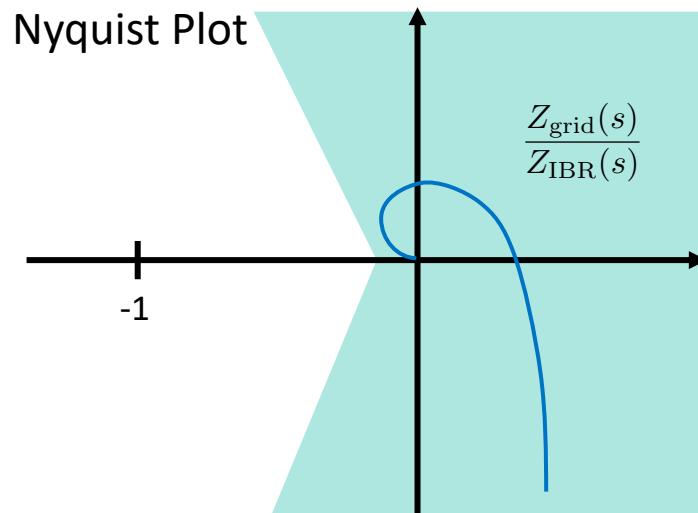
IBR-related Subsynchronous Oscillations

- **When do SSOs occur?**
 - **Weak grids** (e.g. high impedance with GFLs)
 - **Clusters of IBRs in electrically remote areas**
 - Following controller, topology, or operating-point changes
- **What do SSOs depend on?**
 - **Network structure:** line parameters, topology
 - **Control configuration:** GFL vs GFM, PLL dynamics, outer/plant controllers,
 - **Operating point:** voltage setpoints, generation dispatch, the resulting power flow patterns

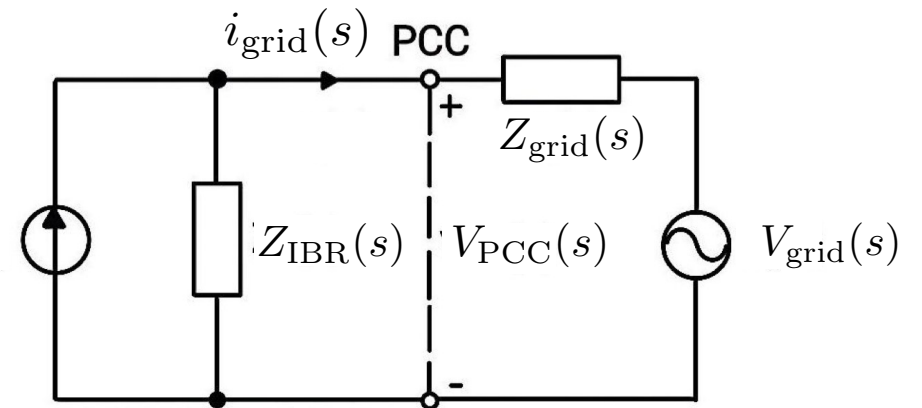
Broader goal: Scalable assessment of IBR-related small-signal stability under changing grid conditions.

Conventional Impedance-Based Stability Analysis

- Impedance models **can capture SSOs**
 - At the Point of Interconnection, stability can be analyzed by comparing inverter and grid impedances.
 - **Nyquist loop-gain criterion** $L(s) = \frac{Z_{\text{grid}}(s)}{Z_{\text{IBR}}(s)}$ explains why weak grids (high Z_{grid}) are more prone to instability.

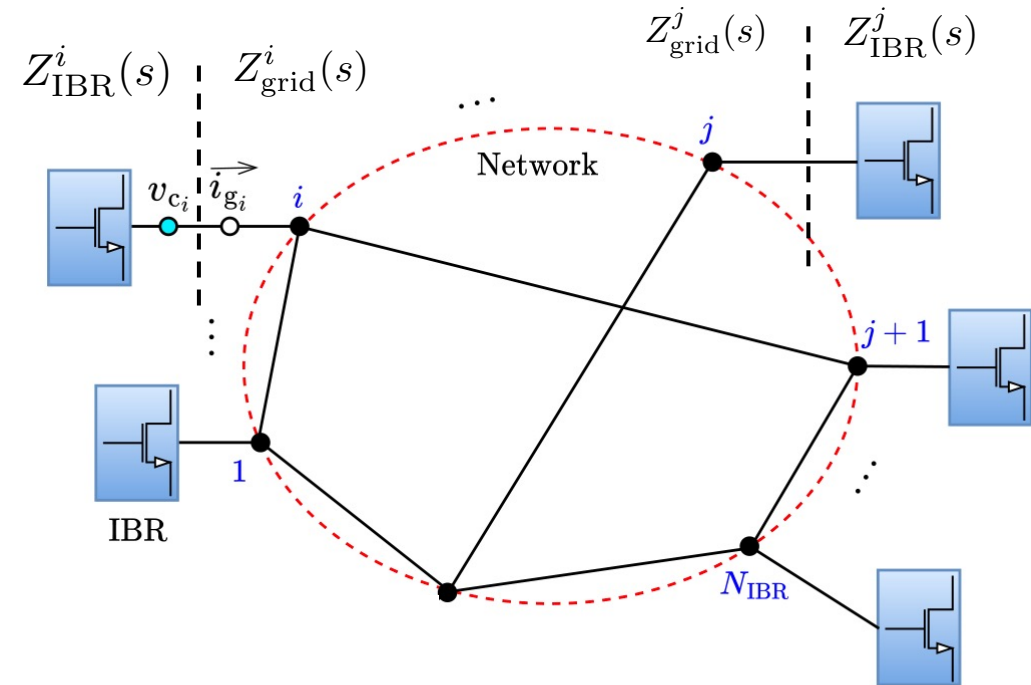


$$V_{\text{PCC}}(s) = \frac{1}{1 + \frac{Z_{\text{grid}}(s)}{Z_{\text{IBR}}(s)}} V_{\text{grid}}(s).$$



Challenges of Impedance Stability Analysis

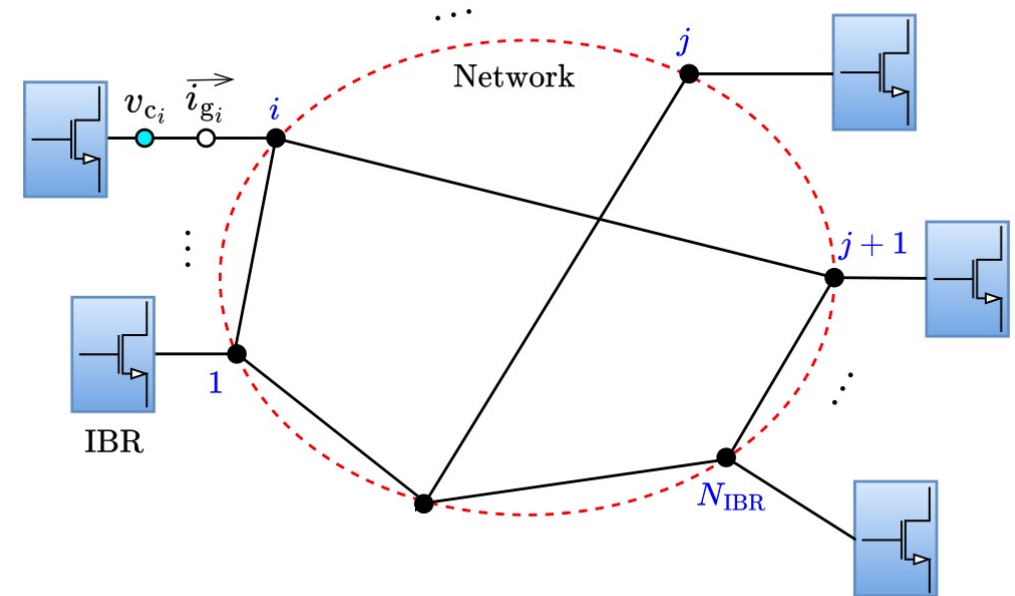
- Z_{IBR}^i depends on:
 - Vendor technology
 - Setpoints (P_i, Q_i)
- Z_{grid}^i depends on:
 - Network topology
 - Power flows (P_{net}, Q_{net})
 - Other connected devices



$$Z_{grid}^i(s) \neq Z_{grid}^j(s)$$

Robust, Decentralized, Small-Signal Analysis

- **Goal:** Develop small-signal stability analysis methods that account for **IBR dynamics** & **network operating conditions**.
- **A useful framework should:**
 - Support device-level assessment using *local* models or measurements
 - *Explicitly* account for network structure and operating point
 - Identify certified stable operating regions for controller tuning and dispatch



Related Literature

Work	Coordinates	Operating Conditions	Info Requirements	Other Properties
[1]	dq-voltage and current: $v^{d/q}, i^{d/q}$	$\theta_i \approx \theta_j, v_i \approx v_j $, or implicit	local	IQC based
[2]	real power and freq: P, ω	decoupled power flows	local	IQC adjacent, exploits model structure
[3], [4]	dq-voltage and current $v^{d/q}, i^{d/q}$	implicit	distributed	passivity, Port Hamiltonian, microgrids
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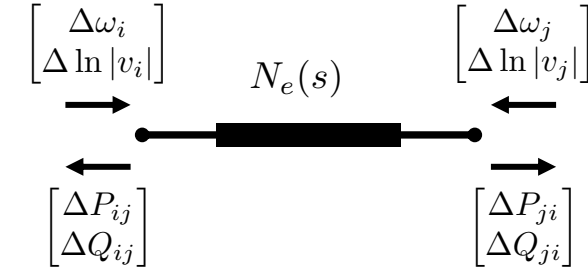
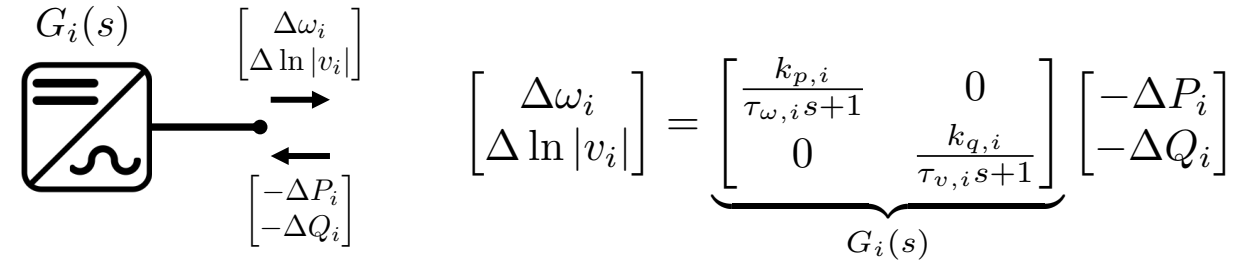
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[12] J. Niehues, R. Delabays, A. Büttner, and F. Hellmann, *Small-signal stability of power systems with voltage droop*, IEEE Trans. Power Systems, 2025.

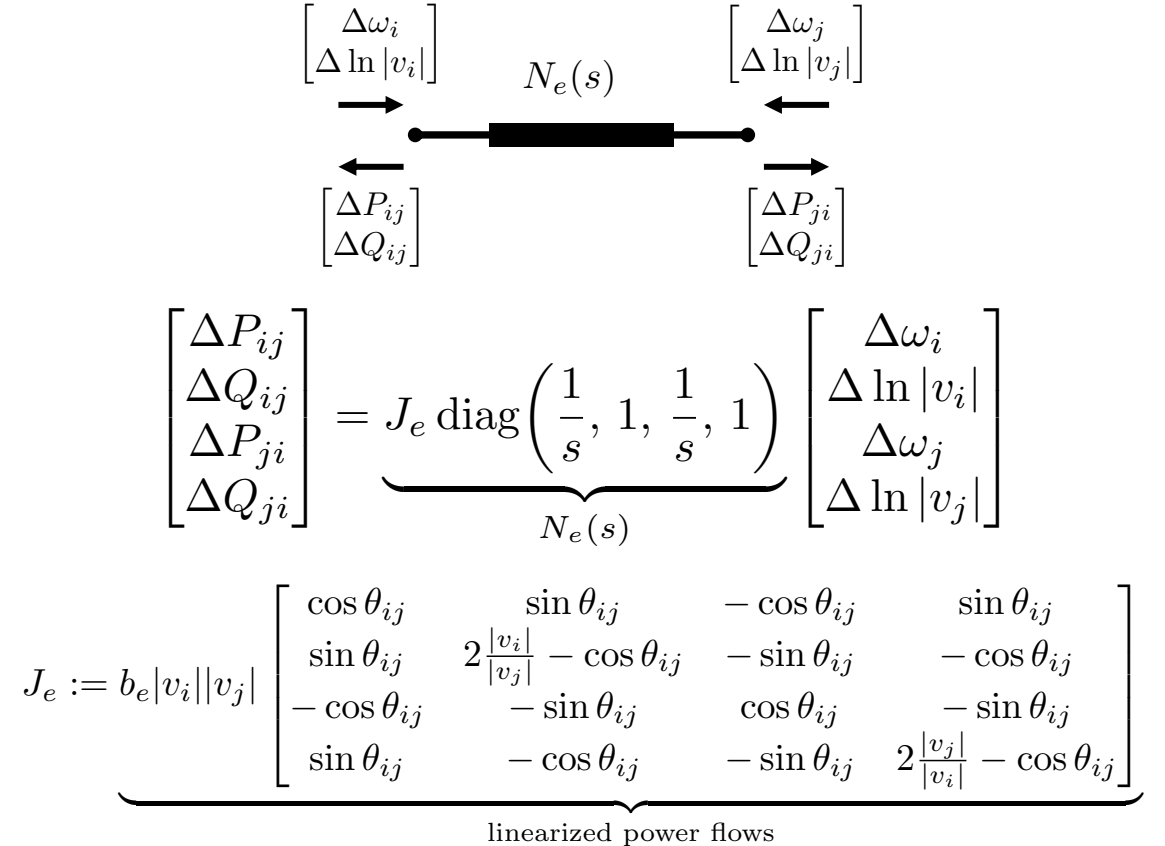
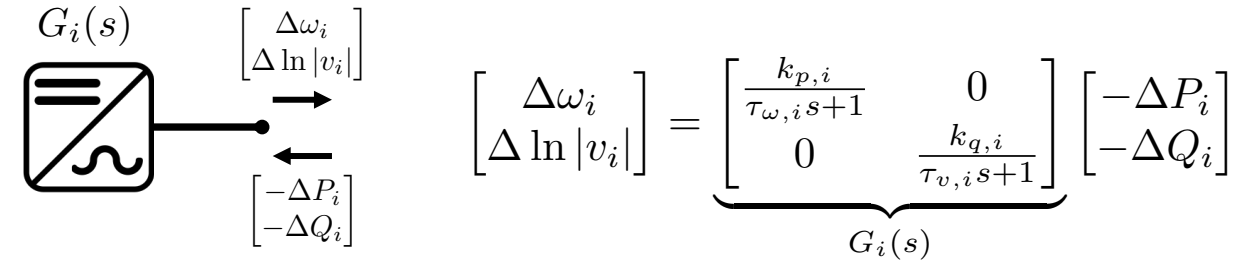
Port-based Network Model

- **IBR Model:** $G_i(s)$
 - GFM converter with filtered droop control
 - Fast inner-loop dynamics are neglected
 - Linearized power and log-polar voltage coordinates: $\Delta P_i, \Delta Q_i, \Delta \ln|v_i|, \Delta \theta_i$

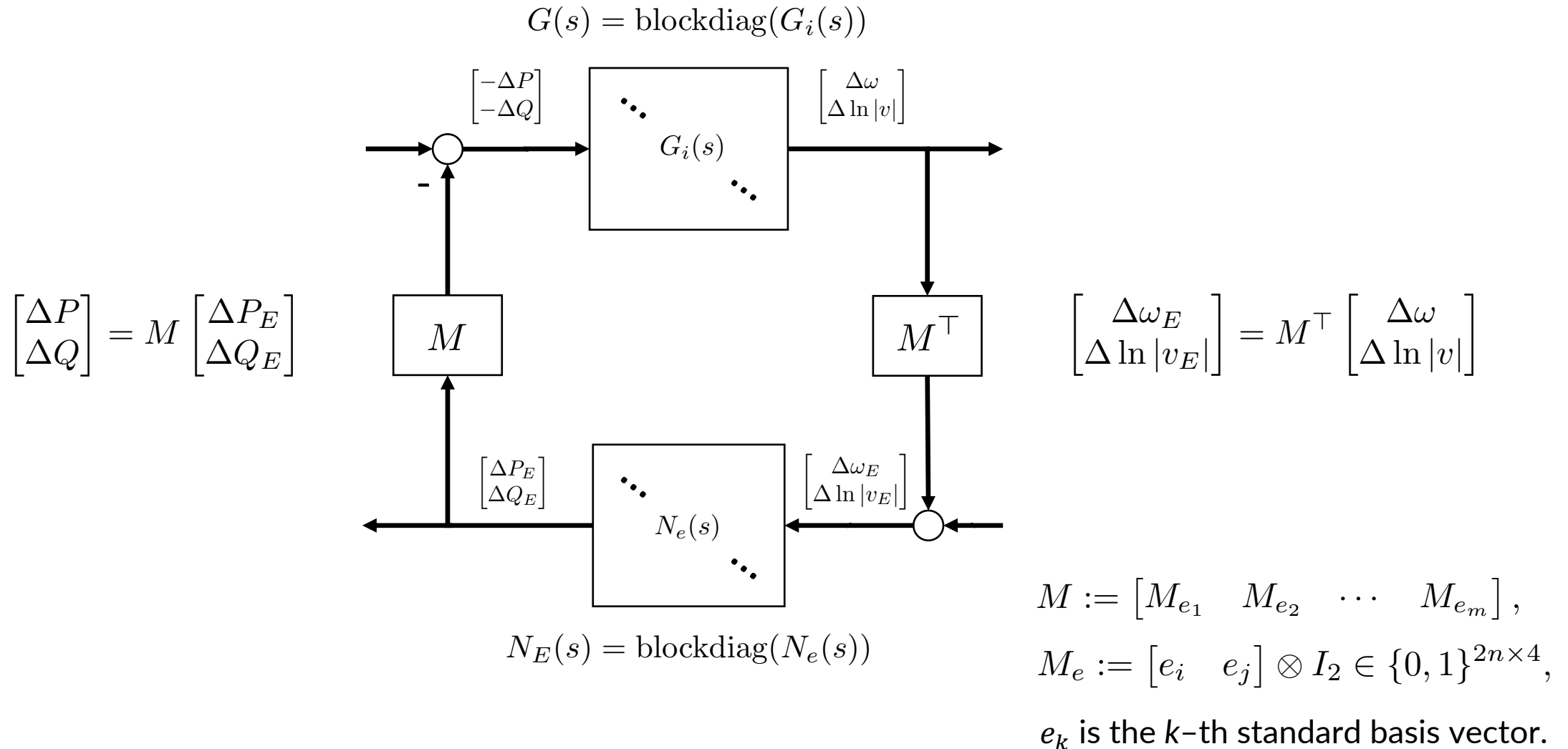


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 - GFM converter with filtered droop control
 - Fast inner-loop dynamics are neglected
 - Linearized power and log-polar voltage coordinates: $\Delta P_i, \Delta Q_i, \Delta \ln|v_i|, \Delta \theta_i$
- **Transmission System:** $N_e(s)$, $e: = \{i, j\}$
 - Lossless inductive lines $b_e = 1/L_e \omega_0$
 - Linearized power and log-polar voltage coordinates: $\Delta P_{ij/ji}, \Delta Q_{ij/ji}, \Delta \ln|v_{i/j}|, \Delta \theta_{i/j}$
 - Electromagnetic transients neglected



Network Interconnection: $G(s) \#_M N_E(s)$



State-aware Decentralized Stability Certificate

Theorem (Main result)

The feedback interconnection $G(s) \#_M N_E(s)$ is internally stable if the following holds:

For each line $e = \{i, j\} \in E$, there exists d_e such that

$$\left(2 \frac{d_e}{b_e |v_i| |v_j|} - \frac{2}{\cos \theta_{ij}}\right) \sqrt{\left(\frac{Q_{ij} - Q_{ji}}{b_e |v_i| |v_j|}\right)^2 + 4} + \left(\frac{d_e}{b_e |v_i| |v_j|}\right)^2 - \frac{2}{\cos \theta_{ij}} \frac{d_e}{b_e |v_i| |v_j|} + 4 \geq 0,$$

where $\sum_{e:i \in e} d_e = \frac{1}{k_{q,i}}$.

(line-wise inequalities with bus-wise allocation)

Understanding Stability Certificates

- Main theorem requires

$$\left(2 \frac{d_e}{b_e |v_i| |v_j|} - \frac{2}{\cos \theta_{ij}}\right) \sqrt{\left(\frac{Q_{ij} - Q_{ji}}{b_e |v_i| |v_j|}\right)^2 + 4} + \left(\frac{d_e}{b_e |v_i| |v_j|}\right)^2 - \frac{2}{\cos \theta_{ij}} \frac{d_e}{b_e |v_i| |v_j|} + 4 \geq 0.$$

- Two key quantities
 - Normalized **reactive power imbalance**

$$\frac{Q_{ij} - Q_{ji}}{b_e |v_i| |v_j|} = \frac{|v_i|}{|v_j|} - \frac{|v_j|}{|v_i|}.$$



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- **Line loading**

$$\cos \theta_{ij} = \sqrt{1 - \left(\frac{P_{ij}}{b_e |v_i| |v_j|}\right)^2}$$



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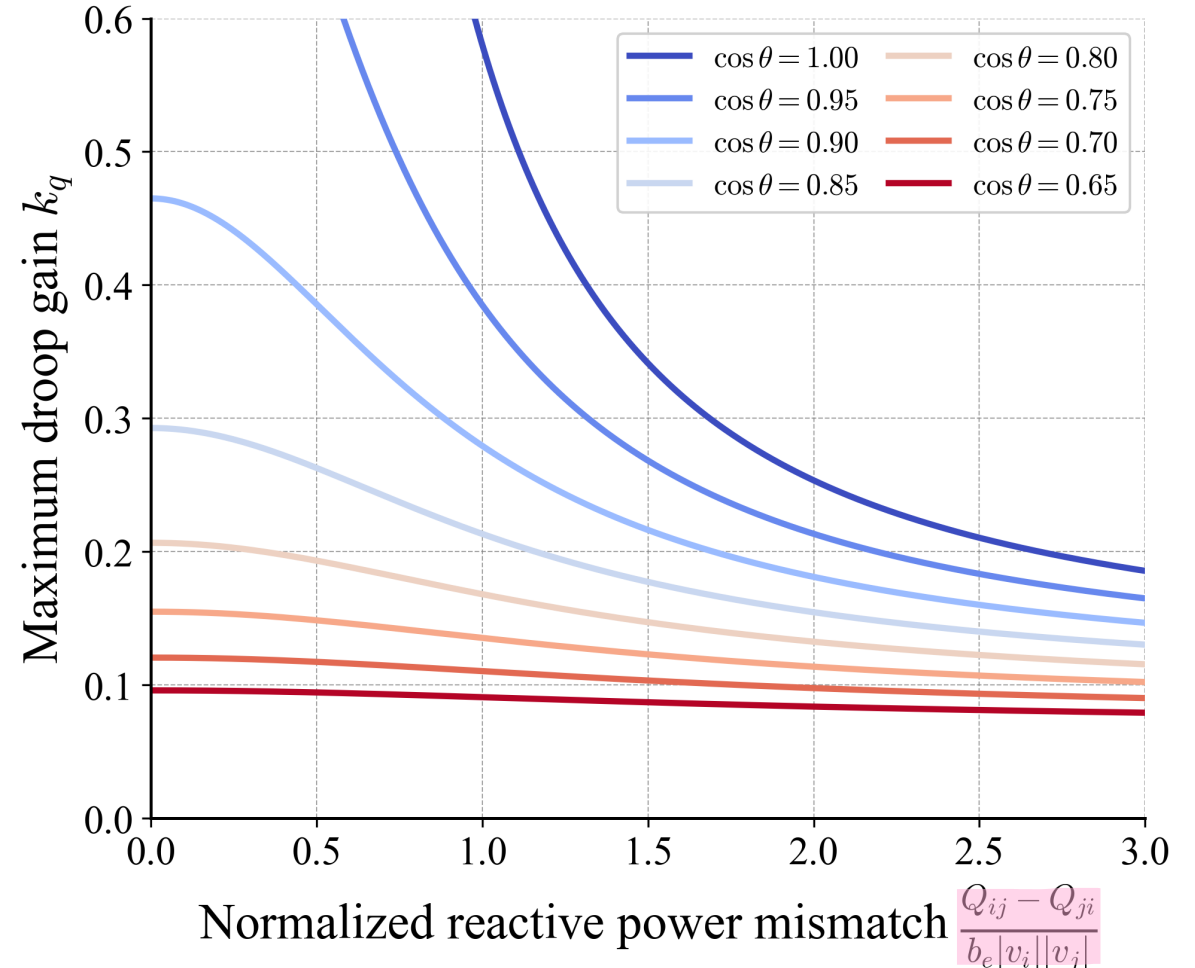
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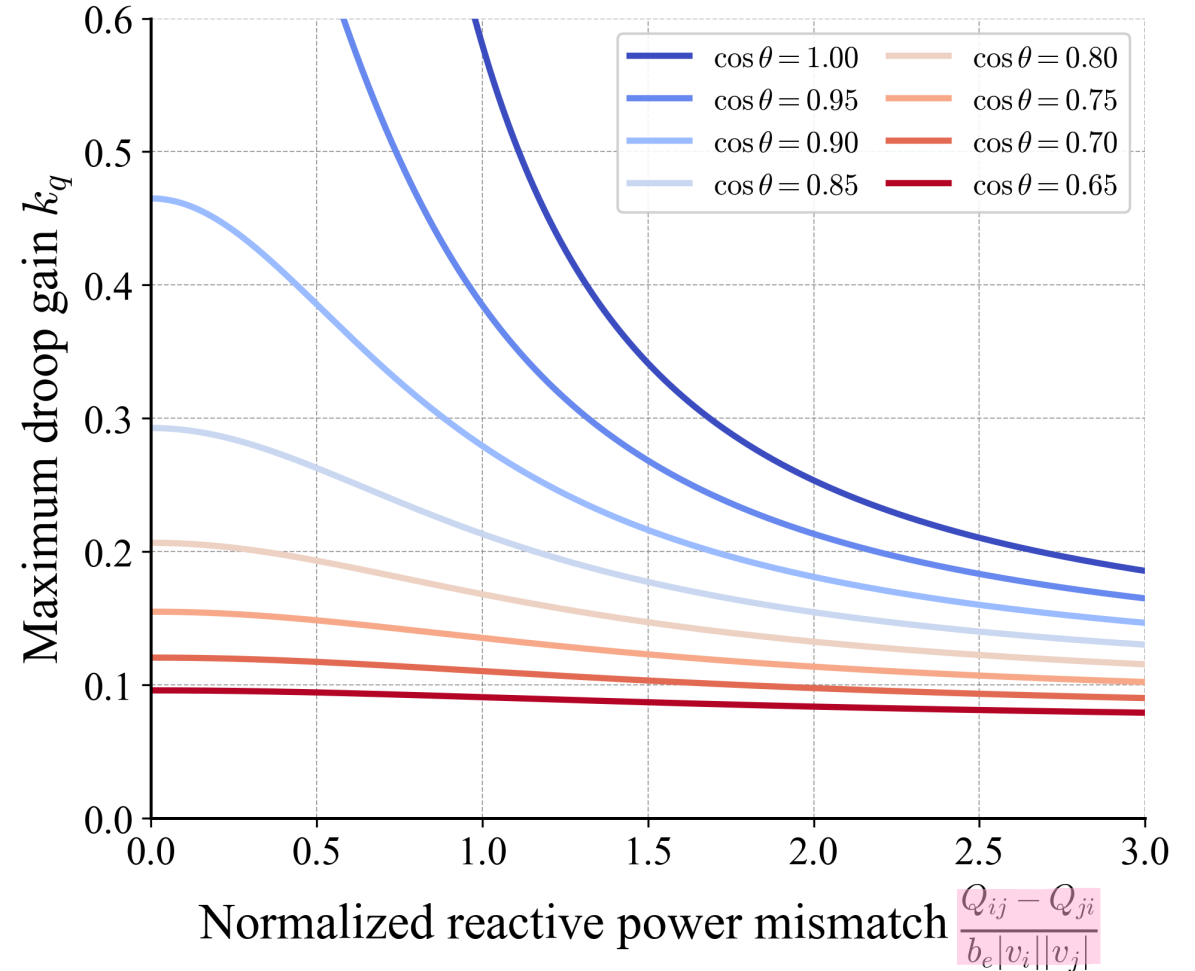
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- Key take-away:

- Larger voltage imbalance and heavier line loading require more conservative reactive droop gains.



Understanding Stability Certificates

- Two key *detrimental* quantities
 - Normalized *reactive power imbalance*

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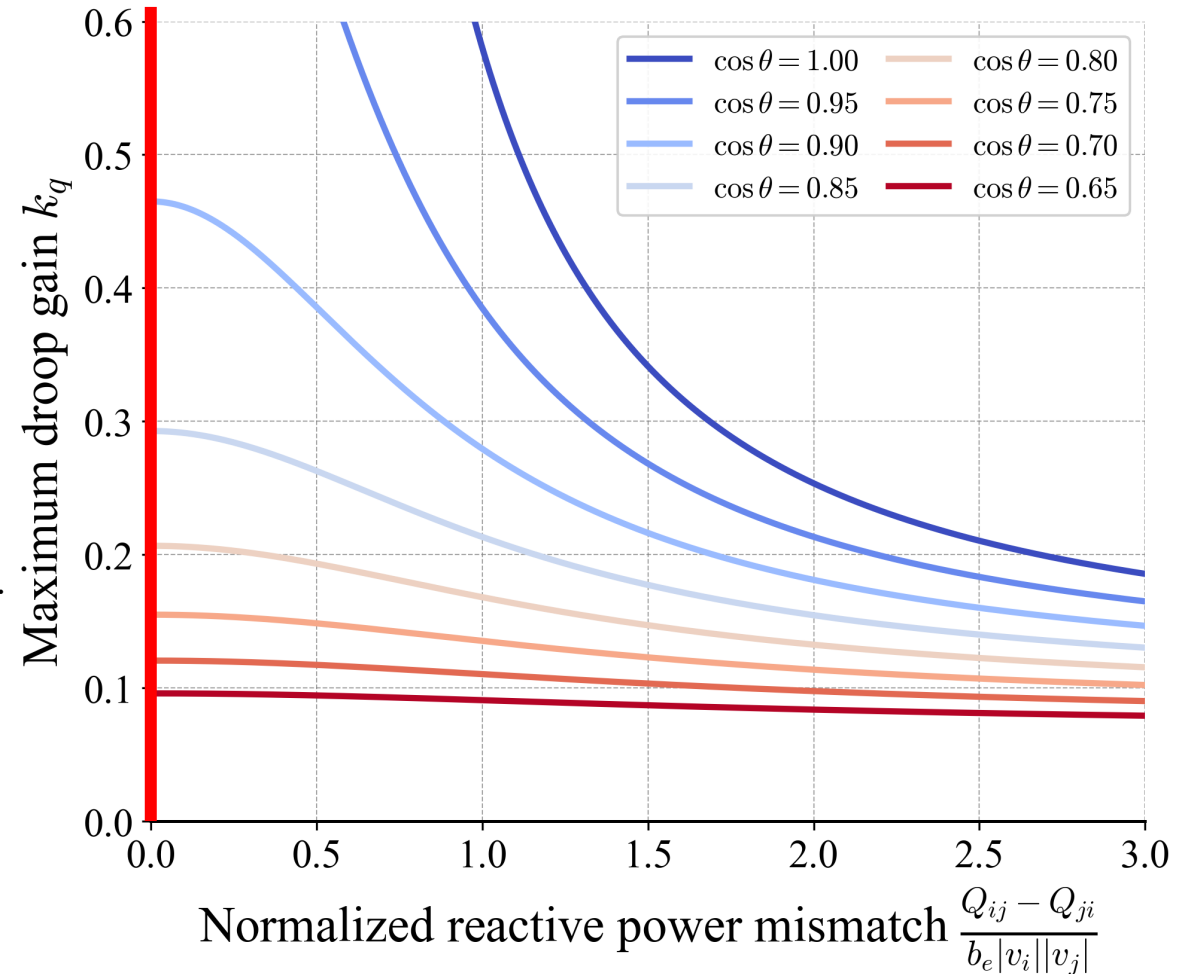
- Line loading*

$$\cos \theta_{ij} = \sqrt{1 - \left(\frac{P_{ij}}{b_e |v_i| |v_j|} \right)^2}$$

- Simplifying assumptions:

a. Equal terminal voltages

$$|v_i| = |v_j| \Rightarrow \max(k_{q,i}, k_{q,j}) < \frac{\cos \theta_{ij}}{2(1 - \cos \theta_{ij}) b_e |v_i| |v_j|}.$$



Understanding Stability Certificates

- Two key *detrimental* quantities
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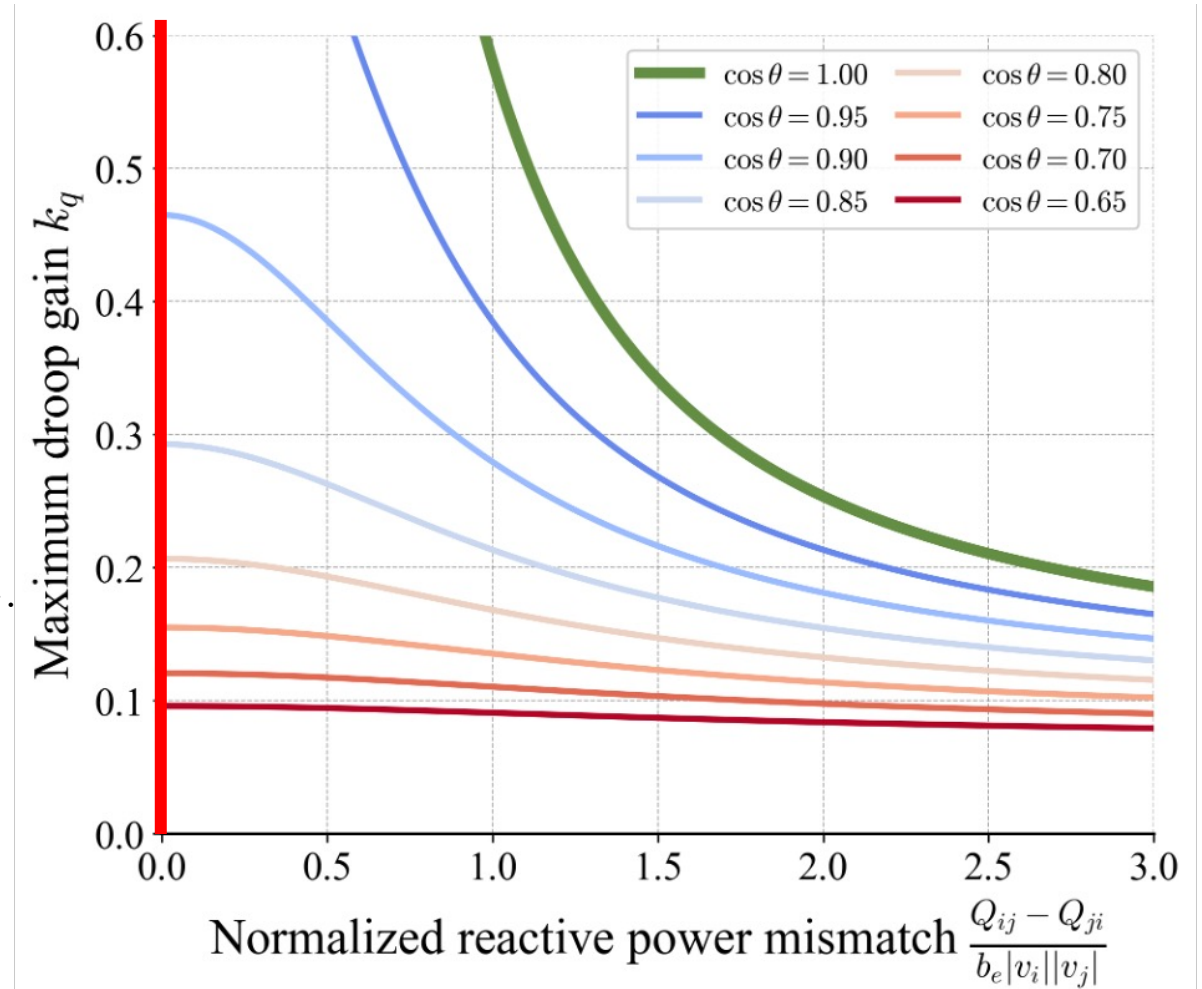
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b. Small phase differences

$$\cos \theta_{ij} = 1 \Rightarrow \max(k_{q,i}, k_{q,j}) < f(|Q_{ij} - Q_{ji}|).$$

a. + b. Zero power flows $\max(k_{q,i}, k_{q,j}) < \infty$.

- Zero power flow o.p.:** certificate satisfied for any finite k_q .
- Other o.p.:** **line loading** and **voltage imbalance** impose a finite, state-dependent gain limit.



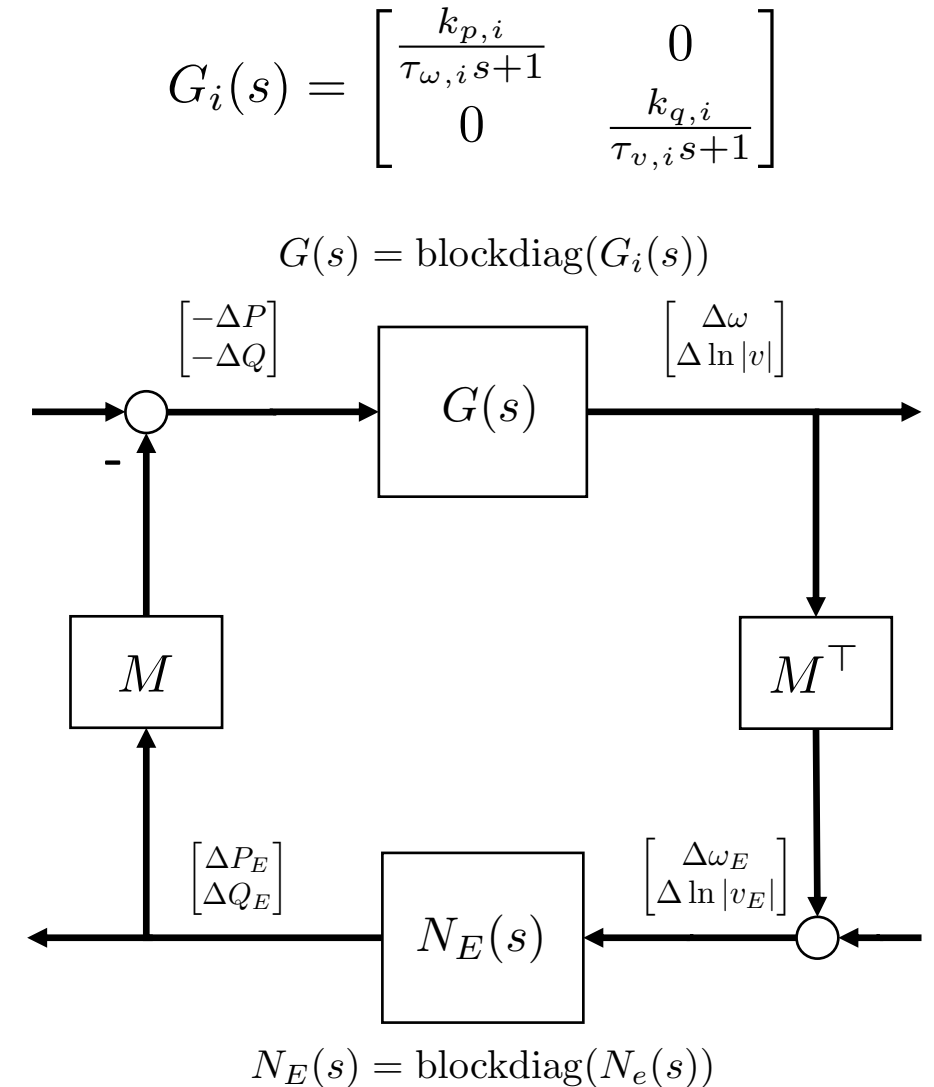
Proof Sketch of Main Result

- **Step 1:** Change voltage coordinates

$$\Delta \ln |v| \rightarrow \Delta \ln \dot{|v|}$$

leading to loop transformation:

$$(i) \quad G(s) \#_M N_E(s) \rightarrow G'(s) \#_M \frac{1}{s} J_E$$



$$N_E(s) = \text{blockdiag}(N_e(s))$$

$$N_e(s) = J_e \text{diag}\left(\frac{1}{s}, 1, \frac{1}{s}, 1\right)$$

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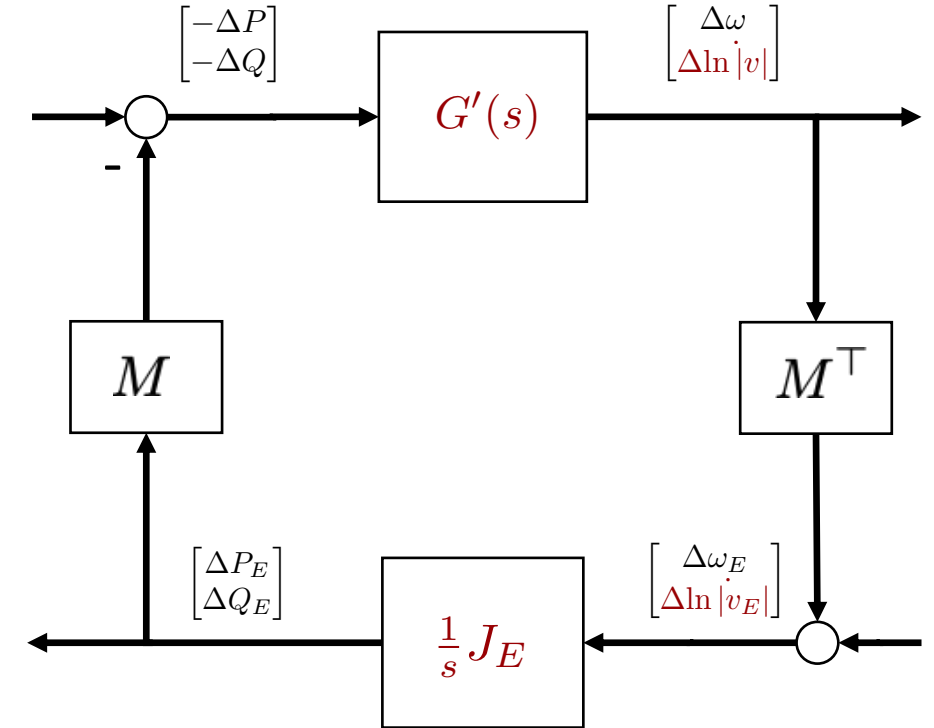
$$J_e + D_e \succeq 0$$

leading to loop transformation:

$$(ii) \quad G'(s) \#_M \frac{1}{s} J_E \rightarrow \hat{G}(s) \#_M \hat{N}_E(s)$$

$$G'_i(s) = \begin{bmatrix} \frac{k_{p,i}}{\tau_{\omega,i}s+1} & 0 \\ 0 & \frac{k_{q,i}s}{\tau_{v,i}s+1} \end{bmatrix}$$

$$G'(s) = \text{blockdiag}(G'_i(s))$$



$$J_E = \text{blockdiag}(J_e)$$

$$J_e := b_e |v_i| |v_j| \begin{bmatrix} \cos \theta_{ij} & \sin \theta_{ij} & -\cos \theta_{ij} & \sin \theta_{ij} \\ \sin \theta_{ij} & 2 \frac{|v_i|}{|v_j|} - \cos \theta_{ij} & -\sin \theta_{ij} & -\cos \theta_{ij} \\ -\cos \theta_{ij} & -\sin \theta_{ij} & \cos \theta_{ij} & -\sin \theta_{ij} \\ \sin \theta_{ij} & -\cos \theta_{ij} & -\sin \theta_{ij} & 2 \frac{|v_j|}{|v_i|} - \cos \theta_{ij} \end{bmatrix}$$

linearized power flows

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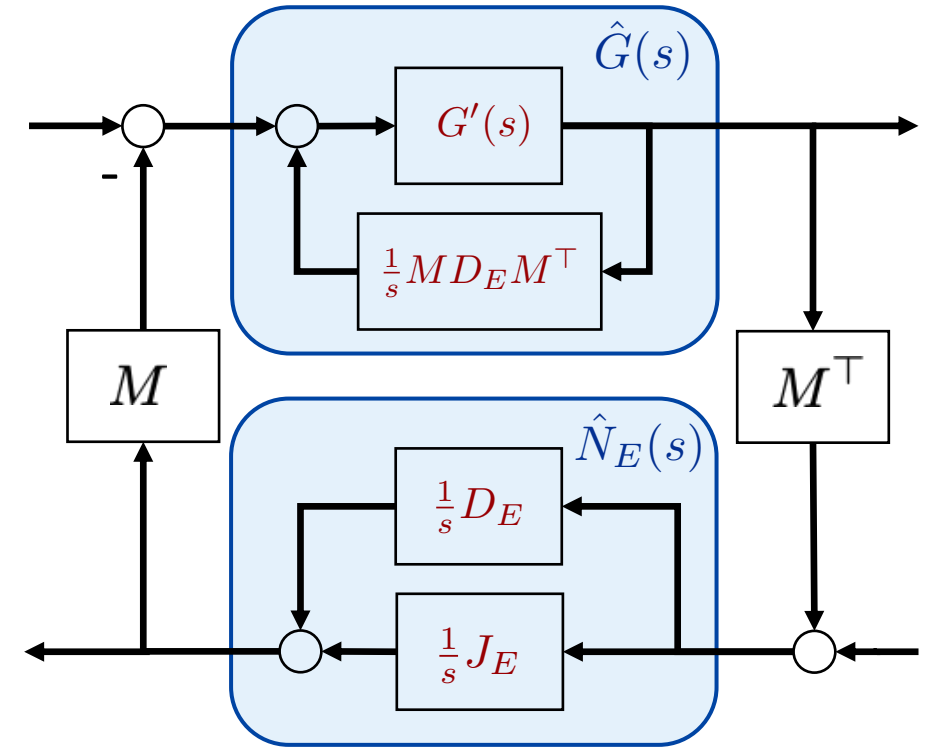
$$(ii) \quad G'(s) \#_M \frac{1}{s} J_E \rightarrow \hat{G}(s) \#_M \hat{N}_E(s)$$

- **Step 3:** Int. Stab. of $\hat{G}(s) \#_M \hat{N}_E(s)$

- **Step 4:** Connect back to original system

$$\hat{G}(s) \#_M \hat{N}(s) \xrightarrow{\text{I.S.}} G'(s) \#_M \frac{1}{s} J_E \xrightarrow{\text{I.S.}} G(s) \#_M N_E(s)$$

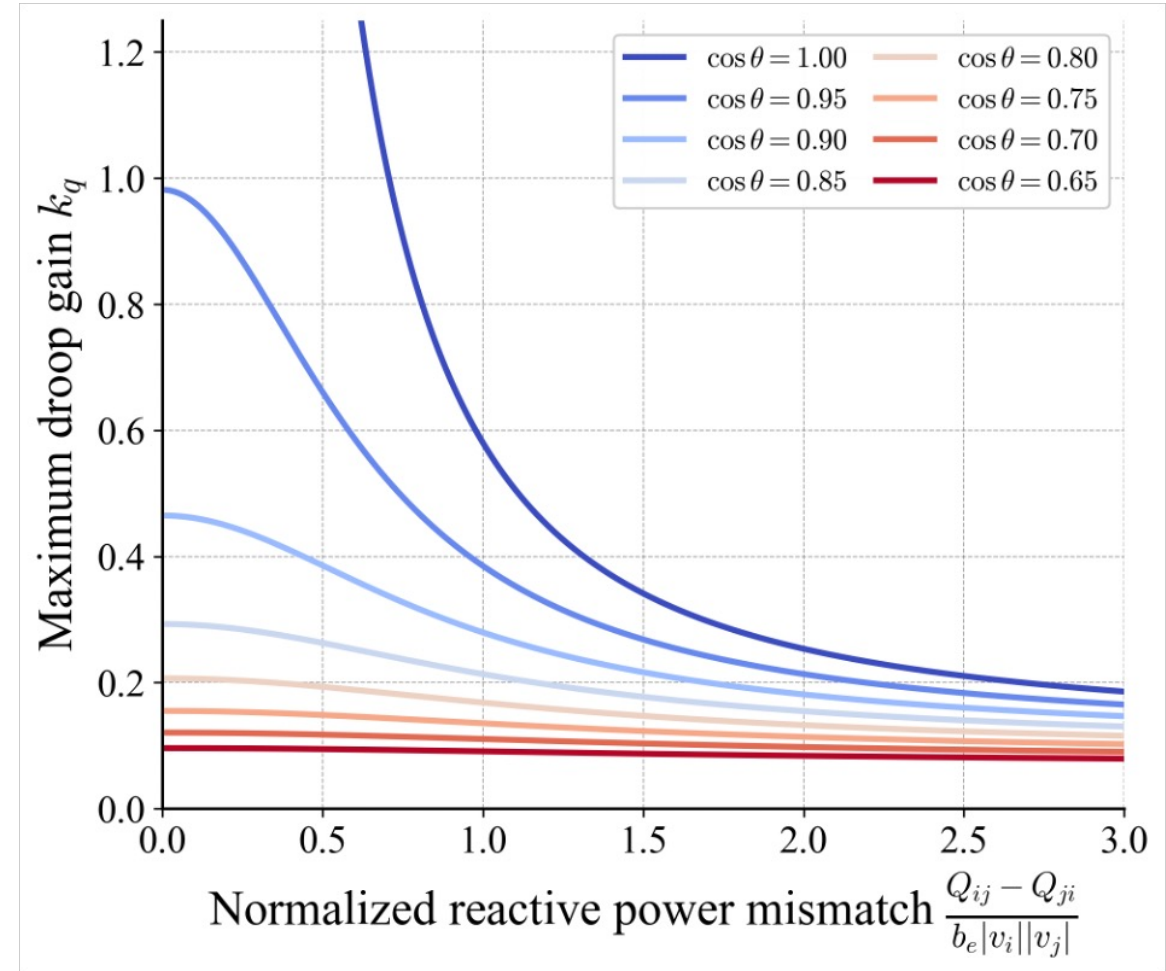
$$G'_i(s) = \begin{bmatrix} \frac{k_{p,i}}{\tau_{\omega,i}s+1} & 0 \\ 0 & \frac{k_{q,i}s}{\tau_{v,i}s+1} \end{bmatrix}$$



$$D_E = \text{blockdiag}(D_e) \quad D_e = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & d_e & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & d_e \end{bmatrix}$$

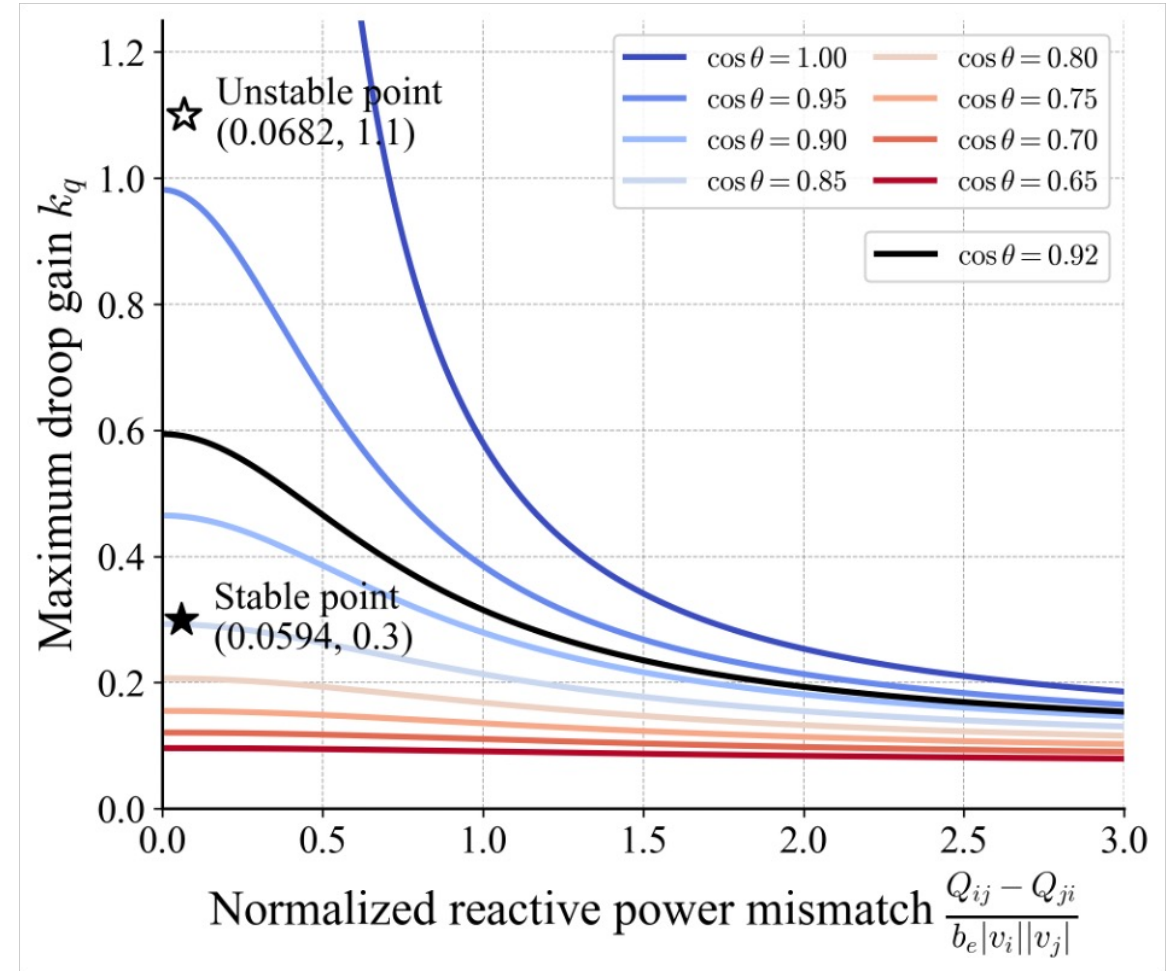
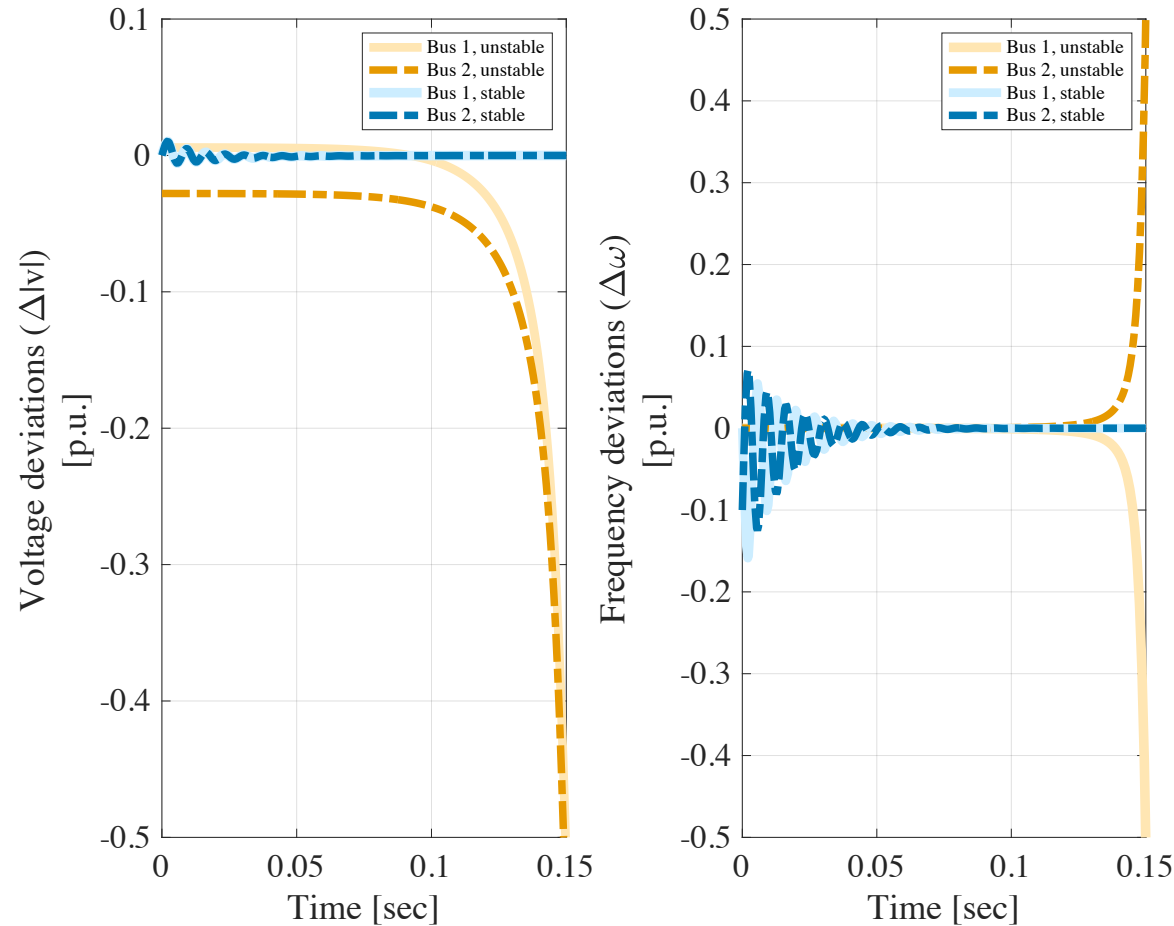
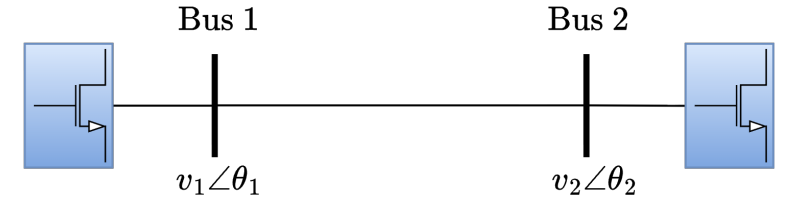
Numerical Illustration

- Operating point: $\cos \theta_{ij} = 0.92$, $\frac{Q_{ij} - Q_{ji}}{b_e |v_1| |v_2|} = 0.059$.
- Control gains: ★ $(k_{q,i}, k_{q,j}) = (0.3, 0.3)$ ☆ $(k_{q,i}, k_{q,j}) = (1.1, 1.0)$



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Summary

Conclusions

- We derived **decentralized stability conditions** that account for the **role of the network state** in small signal stability.
- The resulting stability certificates are **local**, linking controller tuning directly to measurable network quantities.
- Analysis makes explicit the **importance of line loading** and **reactive power mismatch** on stability conditions and tighten admissible voltage-droop gains.
- Stability certificates can be **readily integrated in dispatch** and enable **state-aware IBR tuning**.

Extensions and Future Work

- Broader models: Networks with heterogeneous R/X ratios, heterogeneous GFM/GFL dynamics
- Beyond stability: Performance specification
- Operational integration: Stability-constrained dispatch